



TECHNICAL UNIVERSITY

OF CLUJ-NAPOCA, ROMANIA

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

MULTI-CHANNEL BIOPOTENTIAL DAQ WITH ROS2, IIO BINDINGS

DIPLOMA THESIS

Author: **Ioan Dragomir**

Scientific supervisor: **Șl. Dr. Ing. Tassos Natsakis**

2024



TECHNICAL UNIVERSITY
OF CLUJ-NAPOCA, ROMANIA
FACULTY OF AUTOMATION AND COMPUTER SCIENCE

Approved,

DEAN

Prof.dr.ing. Vlad MUREȘAN

HEAD OF AUTOMATION DEPARMENT

Prof.dr.ing. Honoriu VĂLEAN

Author: Ioan Dragomir

Multi-channel biopotential DAQ with ROS2, IIO bindings

1. **The problem:** Biopotential (ECG, EMG) measurement systems have existed for hundreds of years but current products are by and large closed source, and no previous systems expose their data to the ROS2 or libIIO ecosystems. This thesis develops an open source hardware and software solution for 16 channel 1000Hz ECG and surface EMG measurements, with libIIO and ROS2 bindings on the software side.
2. **Content:** Presentation page, Declaration of authenticity, Summary, Contents, Introduction, Bibliographic research, Analysis design and implementation, Results and validation, Conclusion, Bibliography, Annexes
3. **Place of documentation:** Technical University of Cluj-Napoca
4. **Advisors:**
5. **Thesis emission date:** 02.09.2024
6. **Thesis delivery date:** 04.09.2024

Author signature _____

Scientific supervisor signature _____



TECHNICAL UNIVERSITY
OF CLUJ-NAPOCA, ROMANIA
FACULTY OF AUTOMATION AND COMPUTER SCIENCE

Declaration of authenticity

I, the undersigned, **Ioan Dragomir**, identified by the CI/BI with series RZ, number 113316, CNP 5011113410095, author of the thesis:

Multi-channel biopotential DAQ with ROS2, IIO bindings

conceived for the final graduation examination

at the **Faculty of Automation and Computer Science**,

the Systems Engineering specialization in English,

of the Technical University of Cluj-Napoca,

2023-2024 September session,

declare on my own responsibility that this thesis is the result of my own effort and intellectual activity, based on my own research and on information obtained from the sources cited both in the text and in the bibliography.

I declare that this thesis does not contain plagiarized fragments, and that the bibliographic sources were used in conformity with the Romanian and international legislation regarding ownership rights.

I also declare that this thesis has not been previously submitted to any other university for a higher degree.

In the situation in which this declaration is proven to be false, I will endure all the consequences and administrative sanctions, consisting in the cancellation of the graduation exam.

Date

Ioan Dragomir

02.09.2024



TECHNICAL UNIVERSITY
OF CLUJ-NAPOCA, ROMANIA
FACULTY OF AUTOMATION AND COMPUTER SCIENCE

SUMMARY

of the Diploma Thesis:

Multi-channel biopotential DAQ with ROS2, IIO bindings

Author: **Ioan Dragomir**

Scientific supervisor: **Șl. Dr. Ing. Tassos Natsakis**

1. Requirements: A 8/16 channel ECG and EMG DAQ with at least 1000Hz sample rate, and good signal characteristics (SNR > 20dB, high CMRR, PSRR). Both hardware, firmware and software components designed and implemented. Software bindings for libIIO and ROS2. Open source hardware and software.
2. Solutions: Hardware: MAX78000 microcontroller, AD4114 ADC, custom designed and built pre-amplifier and Driven Right Leg modules. Firmware implements the libIIO protocol, providing a serial iio context. Software provides lightweight bindings for libIIO, Python, and a ROS2 publisher node.
3. Obtained results: All signal quality objectives are checked, with room for improvement for the sample rate when using all 16 channels, and software capabilities.
4. Tests and verification: The hardware components were tested individually using an ADALM2000-based test bench. The system as a whole was tested in various measurement scenarios: single/multiple channel EMG/ECG. The software components' throughput was tested to keep up with ADC output.
5. Personal contribution: Analog signal chain design. Firmware development and low level optimization of third party libraries. Thin software wrappers for interfacing with the rest of the system.
6. Bibliographical sources: M. Barbero, R. Merletti, and A. Rainoldi, "Atlas of Muscle Innervation Zones"; W. J. Esposito, "The ASPEN platform : Tools for mixed signal electronics, digital signal processing, and biomedical electronics"

Author signature _____

Scientific supervisor signature _____